Lecture 8. Import Tariffs and Quotas under Perfect Competition

Lectures on International Trade

Instructor: Xiaoping Chen

Harbin Institute of Technology, Shenzhen

Recall from Previous Lectures and Introduction

- Different models of trade
- Gains from trade
- Distributional effects of trade
- Trade costs
- What is the role for government in international trade?

Outline

- Import Tariffs and Quotas under Perfect Competition
  - World Trade Organization (WTO)
  - The Gains from Trade
  - Import Tariffs for a Small Economy
  - Import Tariffs for a Large Country
  - Import Quotas
- Import Tariffs and Quotas under Imperfect Competition
  - Next Lecture

Introduction

- On September 11, 2009, the U.S. President Barack Obama announced a tariff of 35% on imports of tires made in China.
- The tire tariffs are examples of a **trade policy**, a government action meant to influence the amount of international trade.
- Because the gains from trade are unevenly spread, industries, and labor unions often feel that the government should do something to help limit their losses (or maximize their gains) from international trade.
- That "something" is trade policy, which includes the use of **import tariffs** (taxes on imports), **import quotas** (quantity limits on imports), and **subsidies** for exports.
- Countries can choose their trade policies, following certain agreed international rules.

The World Trade Organization

- After World War II, representatives of the Allied countries met on several occasions to discuss issues such as high trade barriers and unstable exchange rates.
- In 1947 the General Agreement on Tariffs and Trade (GATT) was established. The purpose of which was to reduce barriers to international trade between nations.
- Some of the GATT's main provisions are as follows:
  1. A nation must extend the same tariffs to all trading partners that are WTO members.
  2. Tariffs may be imposed in response to unfair trade practices such as **dumping**.

GATT or WTO's Main Provisions

3 Countries should not limit the quantity of goods and services that they import.

4 Countries should declare **export subsidies** provided to particular firms, sectors, or industries. Article XVI deals with export subsidies, and states that countries should notify each other of the extent of subsidies and discuss the possibility of eliminating them.

5 Countries can temporarily raise tariffs for certain products. Article XIX, called the **safeguard provision** or the escape clause. The importing country can temporarily raise the tariff when domestic producers are suffering due to import competition.

GATT or WTO's Main Provisions, cont'

6 **Regional trade agreements** are permitted under Article XXIV of the GATT.

The GATT recognizes the ability of blocs of countries to form two types of regional trade agreements:

- **Free-Trade Areas**, in which a group of countries voluntarily agree to remove trade barriers between themselves.
- **Customs Unions**, which are free-trade areas in which the countries also adopt identical tariffs between themselves and the rest of the world.

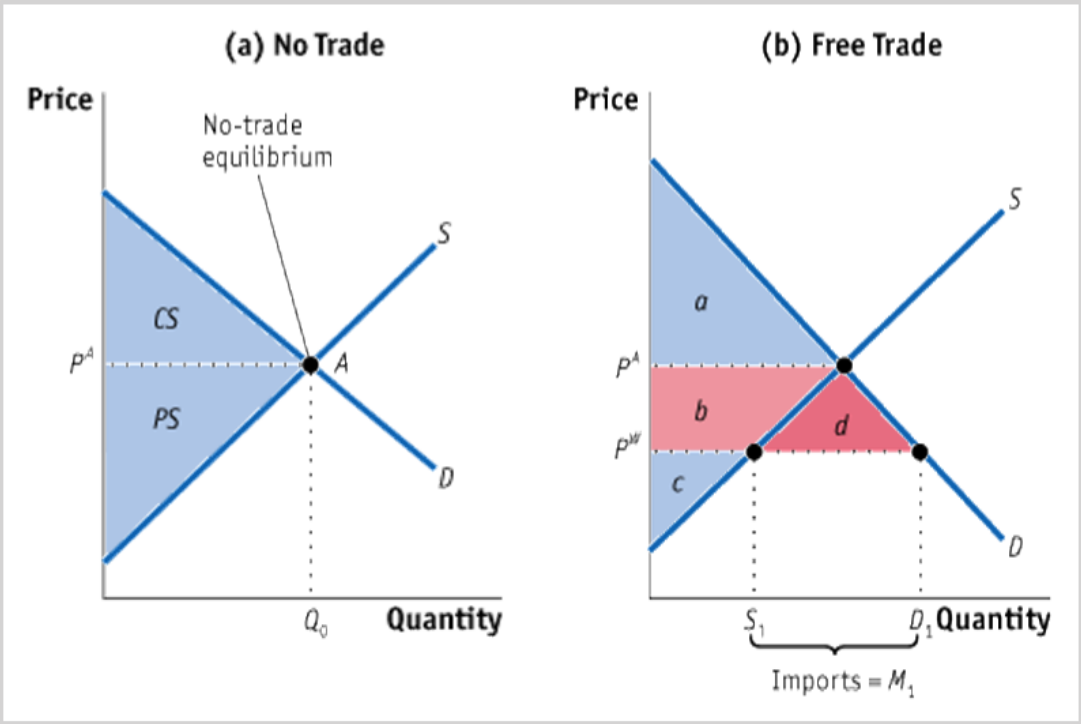
The Gains from Trade

- In previous lectures, we have shown that there are indeed various gains from trade.
  - Higher utility level from indifference curves
  - Higher output level from isoquant curves in offshoring
  - Higher surplus from international movements of labor and capital
- Now we will mainly use the concepts of **consumer surplus** and **producer surplus** to analyze the welfare effects of trade policies.

The Gains from Trade, cont'

No Trade, Free Trade for a Small Country

FIGURE 8-2 The Gains from Free Trade at Home



Rise in consumer surplus: + $(b + d)$

Fall in producer surplus: − b

Net effect on Home welfare: + d

With Home demand of D and supply of S, the no-trade equilibrium is at point A, at the price P^A producing Q_0.

With free trade, the world price is P^W, so quantity demanded increases to D_1 and quantity supplied falls to S_1.

Since quantity demanded exceeds quantity supplied, Home imports $D_1 - S_1$.

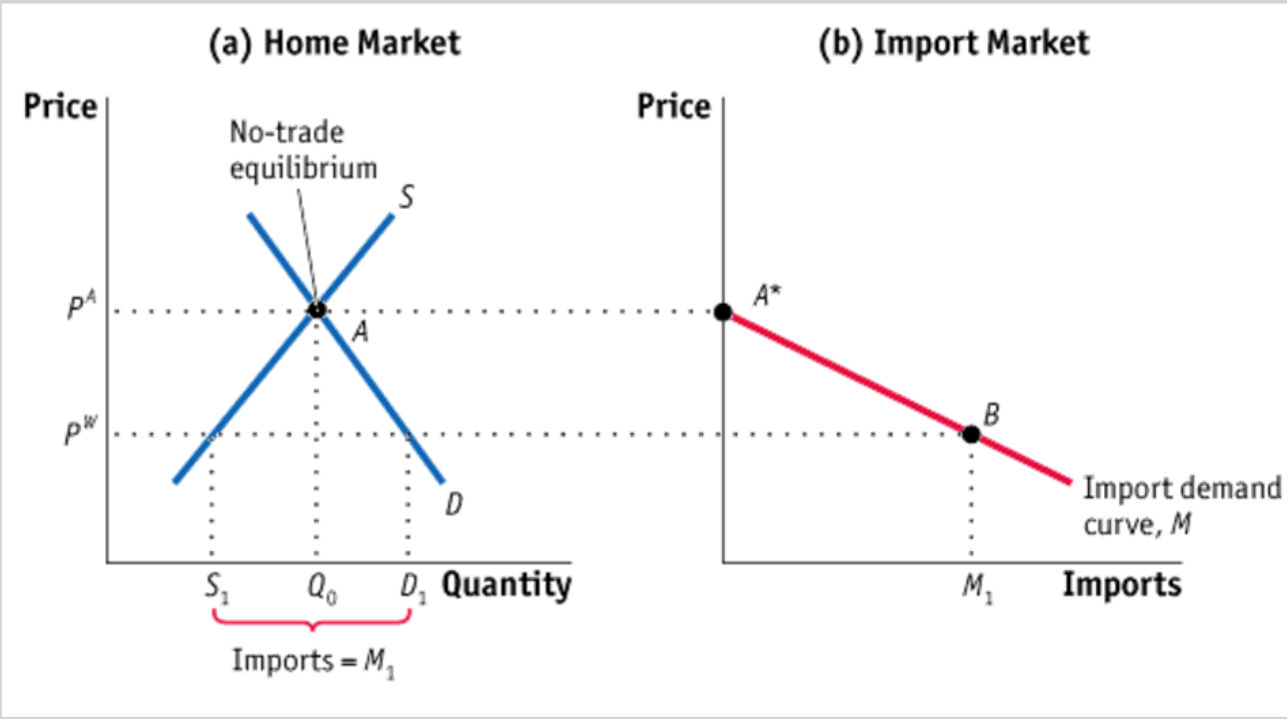
Consumer surplus increases by the area $(b + d)$, and producer surplus falls by area b.

The gains from trade are measured by area d.

Import Tariffs for a Small Economy

Home Import Demand Curve

FIGURE 8-3 Home Import Demand



The **import demand curve** shows the relationship between the world price of a good and the quantity of imports demanded by Home consumers.

With Home demand of *D* and supply of *S*, the no-trade equilibrium is at point *A*, with the price P^A and import quantity Q_0.

Import demand at this price is zero, as shown by the point *A'* in panel (b).

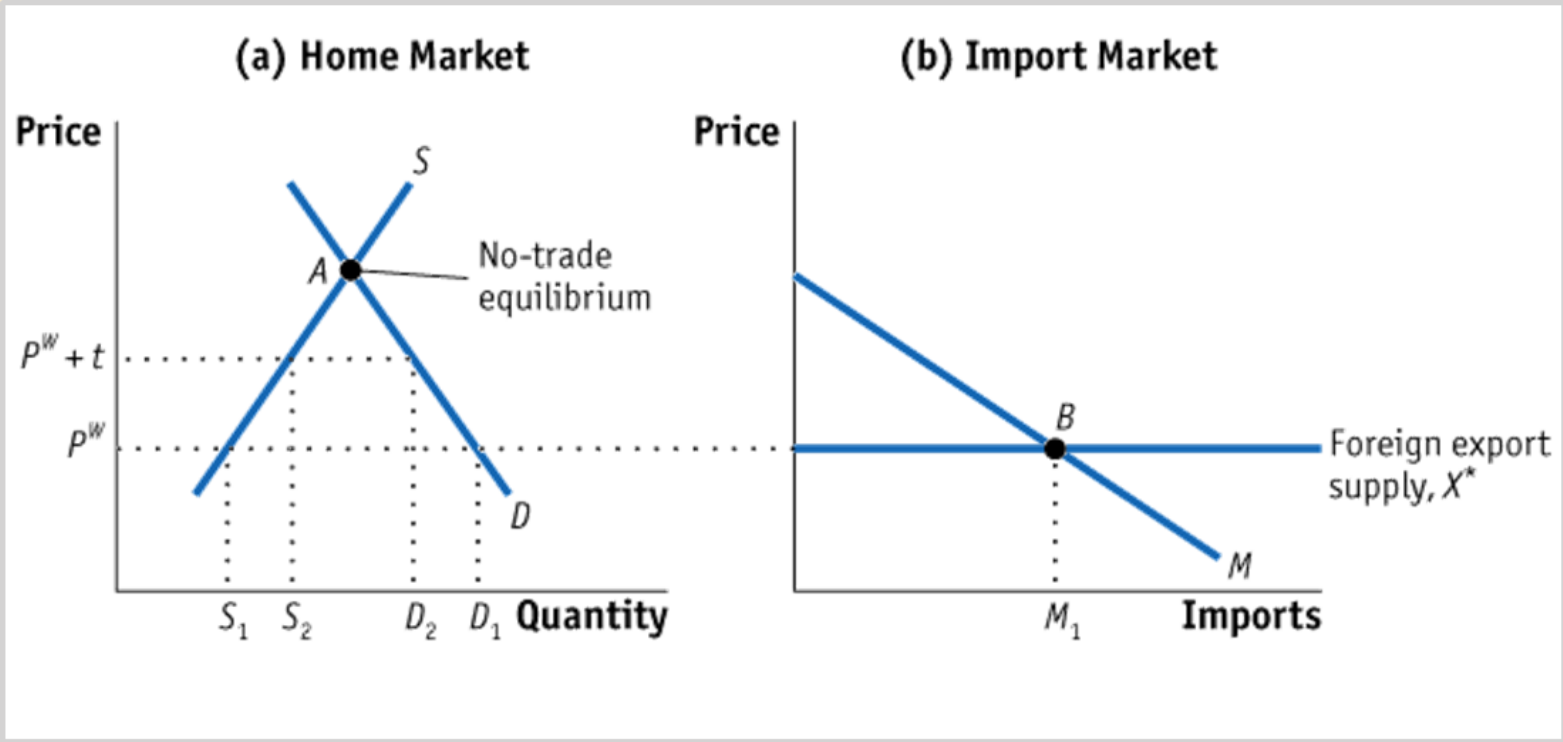
At a lower world price of P^W, import demand is $M_1 = D_1 - S_1$, as shown by point *B*.

Joining up all points between *A'* and *B*, we obtain the import demand curve, *M*.

Import Tariffs for a Small Economy, cont'

Free Trade for a Small Country and Effect of the Tariff

FIGURE 8-4 (1 of 2) **Free Trade for a Small Country**

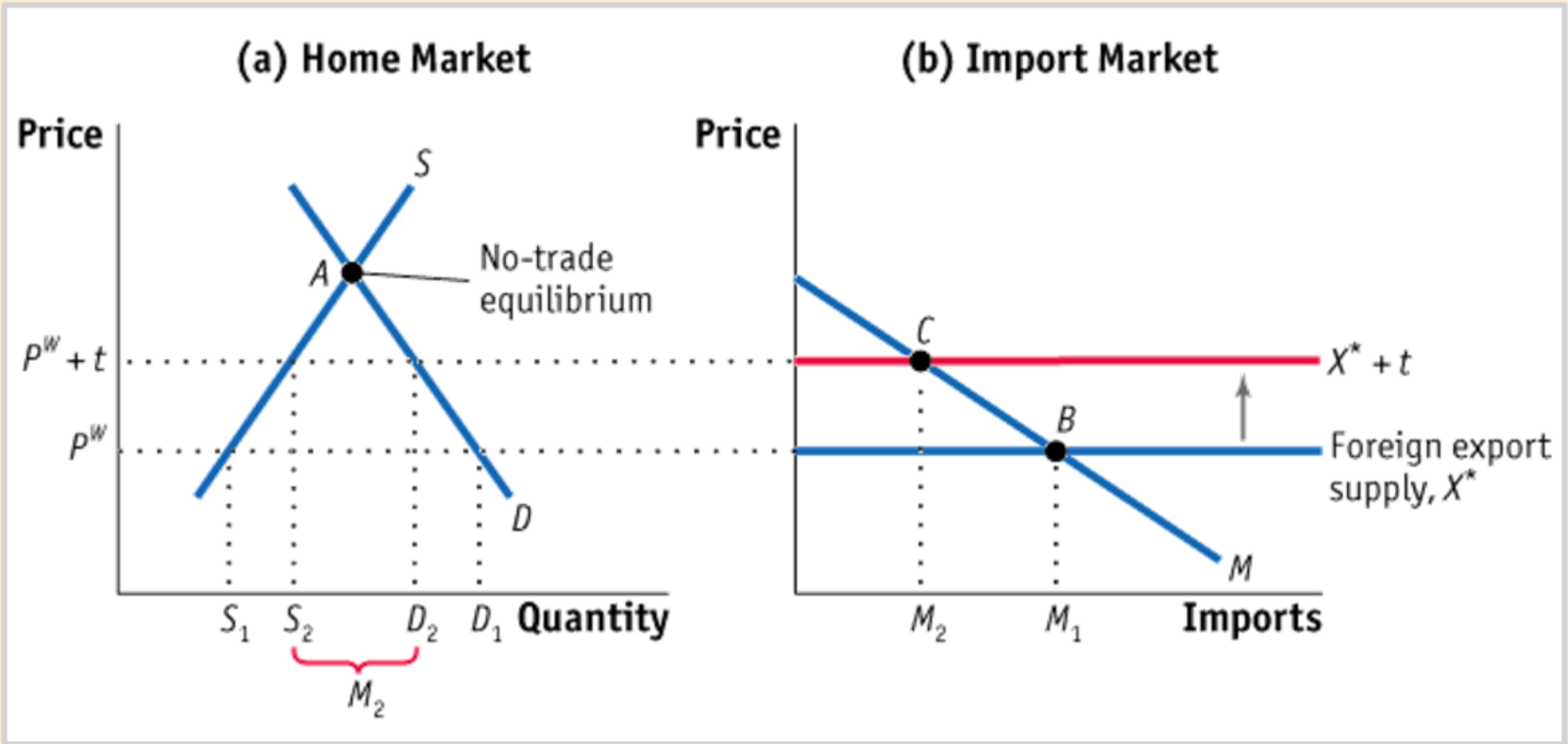


Applying a tariff of *t* dollars will increase the import price from P^W to $P^W + t$. The domestic price of that good also rises to $P^W + t$. This price rise leads to an increase in Home supply from S_1 to S_2, and a decrease in Home demand from D_1 to D_2, in panel (a).

Import Tariffs for a Small Economy, cont'

Free Trade for a Small Country and Effect of the Tariff

FIGURE 8-4 (2 of 2) **Effects of a Tariff**



Tariff for a Small Country

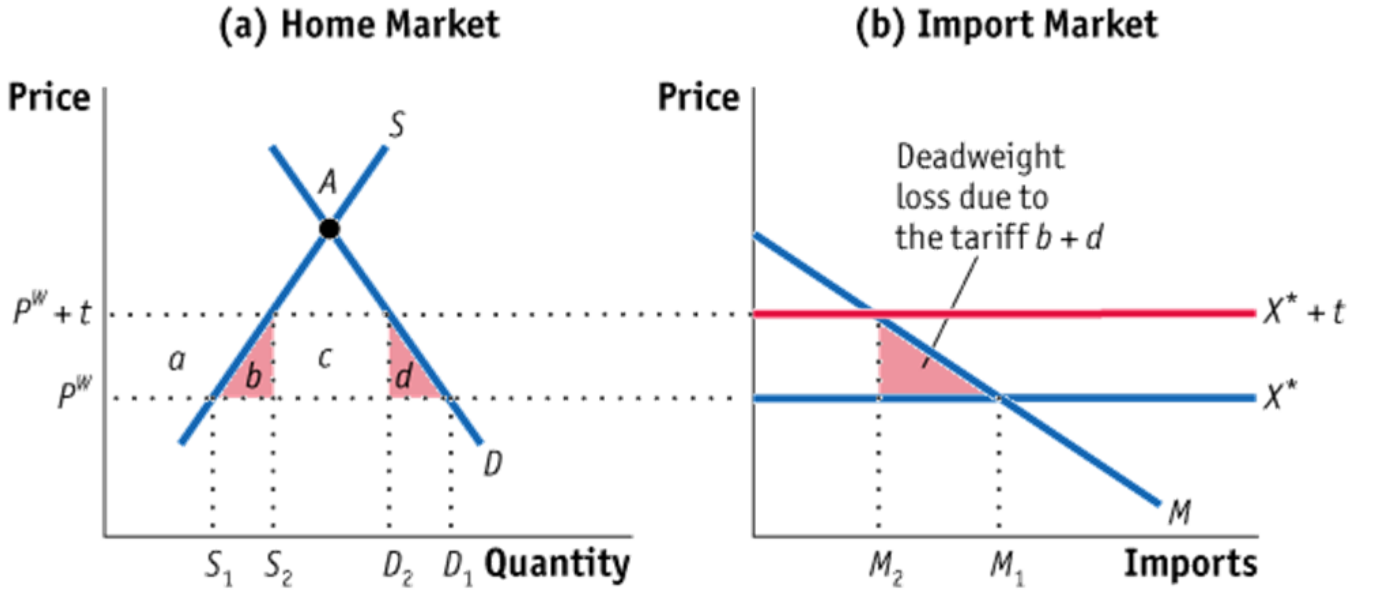
Imports fall due to the tariff, from M_1 to M_2 in panel (b).

As a result, the equilibrium shifts from point *B* to *C*.

Import Tariffs for a Small Economy, cont'

Effects of the Tariff

FIGURE 8-5 Effects of Tariff on Welfare



Fall in consumer surplus: − (*a* + *b* + *c* + *d*)
Rise in producer surplus: + *a*
Rise in government revenue: + c
Net effect on Home welfare: − (*b* + *d*)

The triangle (*b* + *d*) is a **deadweight loss**, or a loss that is not offset by a gain elsewhere in the economy

Therefore, the net loss in welfare, the deadweight loss to Home, is *(b + d)*, which is measured by the two triangles *b* and *d* in panel (a) or the single (combined) triangle *b + d* in panel (b).

Import Tariffs for a Small Economy, cont'

Effects of the Tariff

- The area of triangle b equals the increase in marginal costs for the extra units produced and can be interpreted as the **production loss** (or the *efficiency loss*) for the economy due to producing at marginal cost above the world price.
- The area of the triangle d can be interpreted as the drop in consumer surplus for those individuals who are no longer able to consume the units between D_1 and D_2 because of the higher price. We refer to this drop in consumer surplus as the **consumption loss** for the economy.
- If a small country suffers a loss when it imposes a tariff, why do so many have tariffs as part of their trade policies?

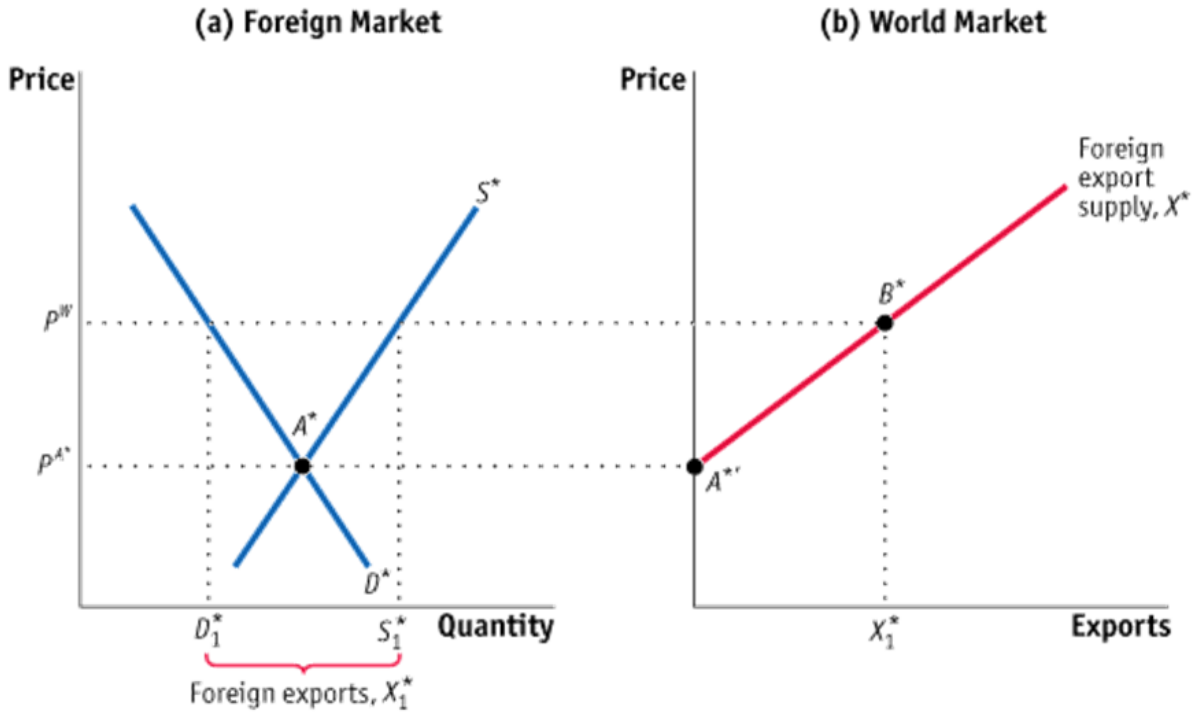
Import Tariffs for a Small Economy, cont'

- One answer is that a developing country does not have any other source of government revenue. Import tariffs are "easy to collect".
- A second reason is politics. The benefits to producers (and their workers) are typically more concentrated on specific firms and states than the costs to consumers, which are spread nationwide.
- The WTO has a formal **dispute settlement procedure** under which countries that believe that the WTO rules have not been followed can bring their complaint and have it evaluated.
- When the U.S. impose tariff on its steel import, the European Union (EU) took action by bringing the case to the WTO. The WTO ruling entitled the EU and other countries to retaliate against the U.S. by imposing tariffs of their own against U.S. exports.
- The use of tariffs by an importer can easily lead to a response by exporters and a **tariff war**.

Import Tariffs for a Large Country

Foreign Export Supply:

FIGURE 8-6 (2 of 3) **Foreign Export Supply**



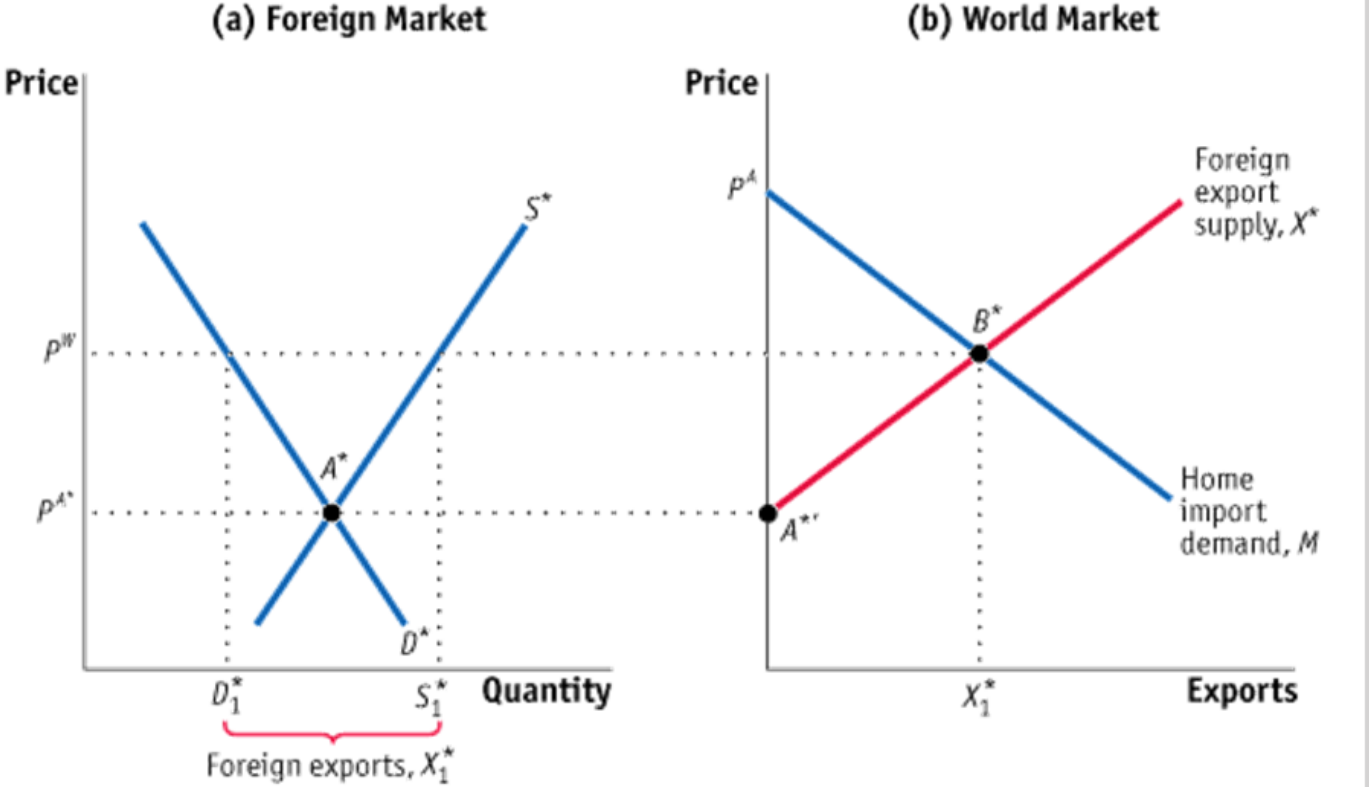
When the world price, P^W, is higher than Foreign no-trade price, the quantity supplied by Foreign, S^*_1, exceeds the quantity demanded by Foreign, D^*_1, and Foreign exports $X^*_1 = S^*_1 - D^*_1$.

In panel (b), joining up points $A^{*\prime}$ and B^*, we obtain the upward-sloping export supply curve X^*.

Import Tariffs for a Large Country, cont'

Foreign Export Supply and Home Demand:

FIGURE 8-6 (3 of 3) **Foreign Export Supply (continued)**

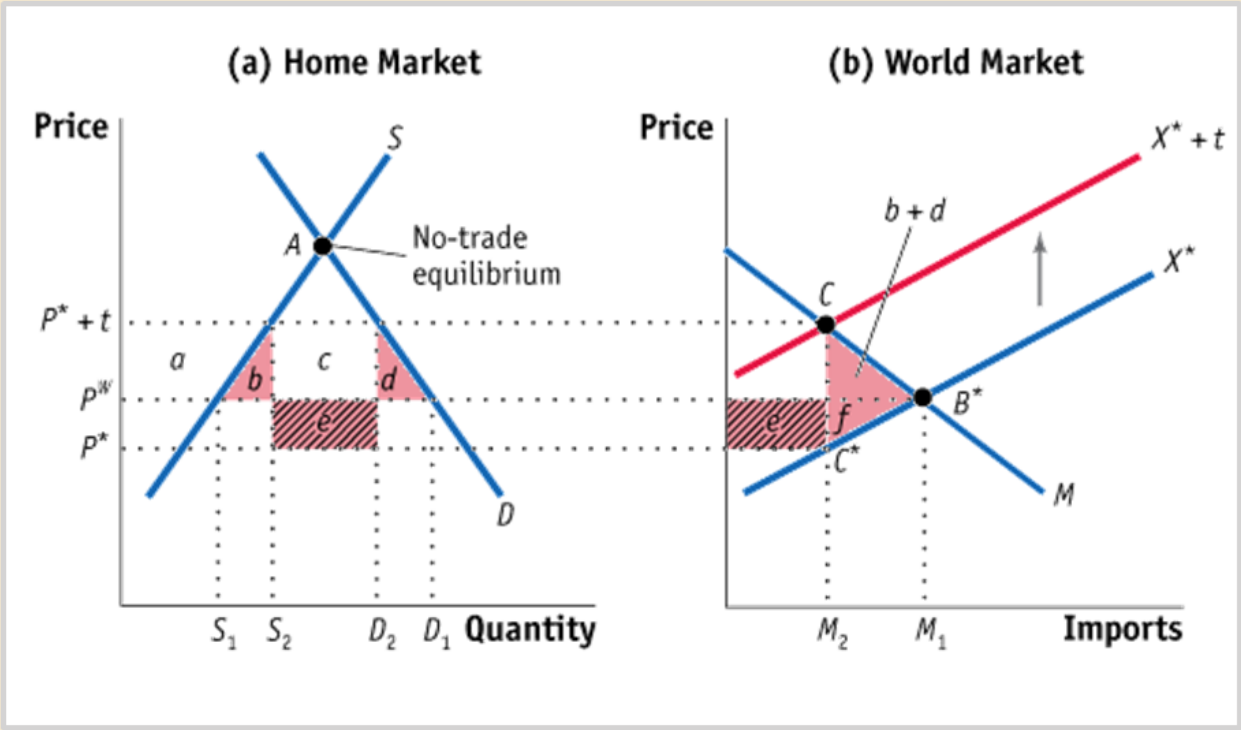


With the Home import demand of *M*, the world equilibrium is at point *B, with the price P^W.**

Import Tariffs for a Large Country, cont'

Effects of the Tariff:

FIGURE 8-7 Tariff for a Large Country



Fall in consumer surplus: − (*a* + *b* + *c* + *d*)

Rise in producer surplus: + *a*

Rise in government revenue: + (*c* + *e*)

Net effect on Home welfare: *e* − (*b* + *d*)

Area *e* is a measure of the **terms-of-trade gain** for the importer.

The tariff shifts up the export supply curve from X^* to $X^* + t$.

As a result, the Home price increases from P^W to $P^* + t$, and the Foreign price falls from P^W to P^*.

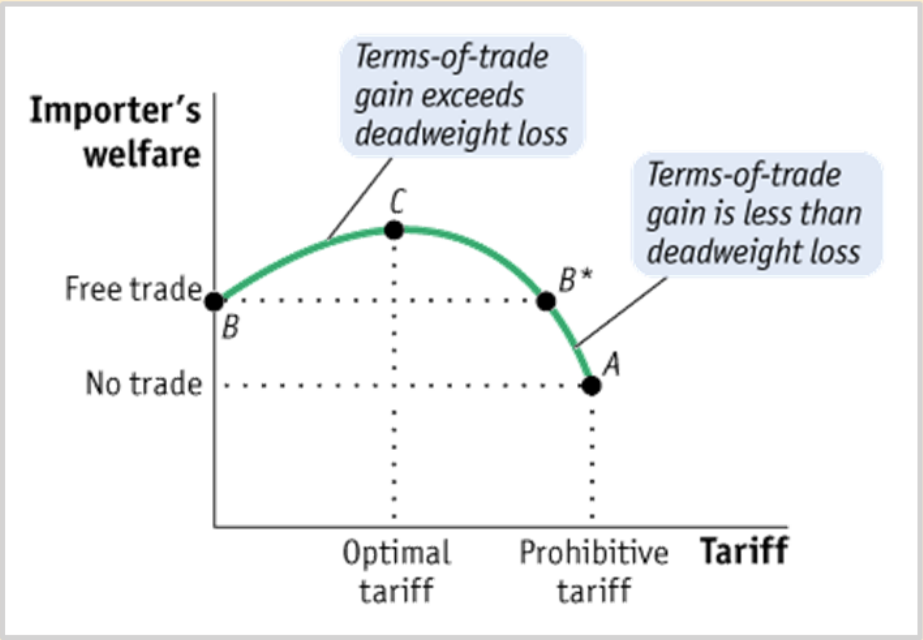
The deadweight loss at Home is the area of the triangle *(b + d)*, and Home also has a terms-of-trade gain of area *e*.

Foreign loses the area *(e + f)*, so the net loss in world welfare is the triangle *(b + d + f)*.

Import Tariffs for a Large Country, cont'

The **optimal tariff** is defined as the tariff that leads to the maximum increase in welfare for the importing country.

FIGURE 8-8 Tariffs and Welfare for a Large Country



Optimal Tariff Formula The formula depends on the elasticity of Foreign export supply, which we call E^*_X.

$$\text{Optimal tariff} = \frac{1}{E^*_X}$$

For a large importing country, a tariff initially increases the importer's welfare because the terms-of-trade gain exceeds the deadweight loss. So the importer's welfare rises from point *B*.

Welfare continues to rise until the tariff is at its optimal level (point *C*).

After that, welfare falls.

If the tariff is too large (greater than at *B*'), then welfare will fall below the free-trade level.

For a prohibitive tariff, with no imports at all, the importer's welfare will be at the no-trade level, at point *A*.

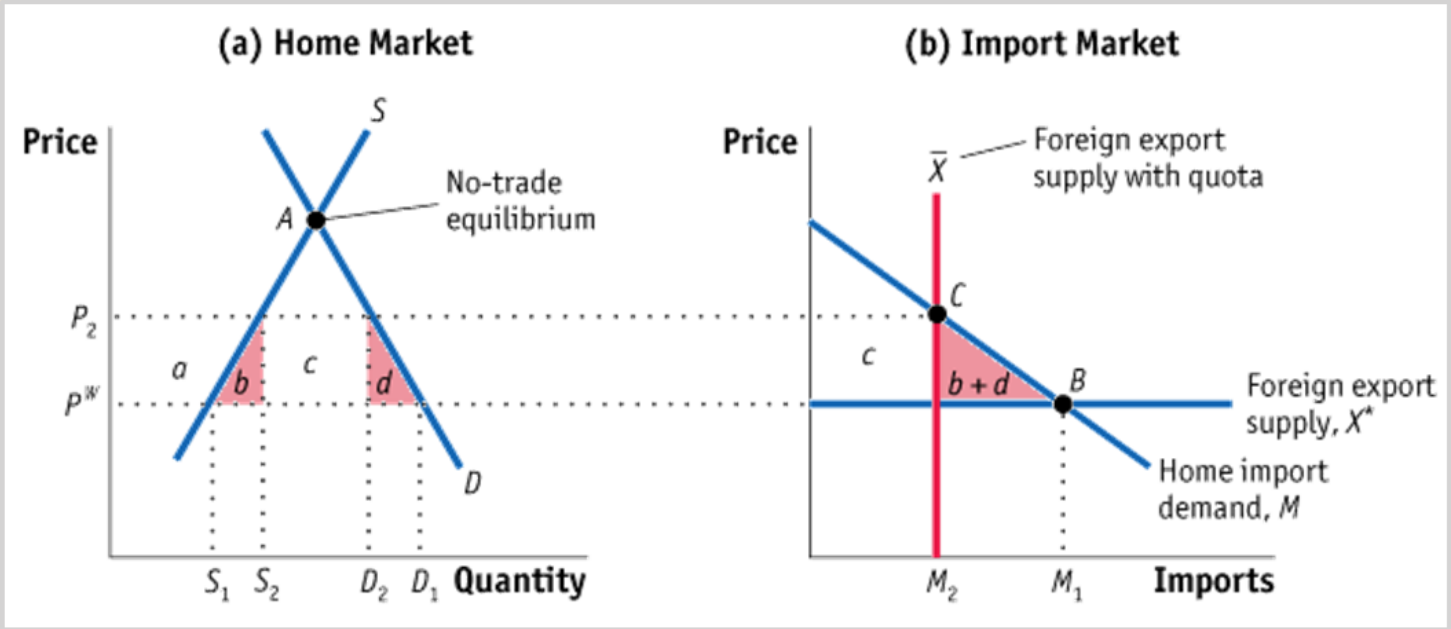
Import Quotas

- On January 1, 2005, China was poised to become the world's largest exporter of textiles and apparel. On that date, a system of worldwide import quotas known as the **Multi-Fibre Arrangement** (MFA) was abolished.
- The agreement imposed quotas on the amount that developing countries could export in the form of yarn, fabric and clothing to developed countries.
- Besides the MFA, there are many other examples of import quotas. For example, since 1993 Europe had a quota on the imports of bananas that allowed for a greater number of bananas to enter from its former colonies in Africa than from Latin America.
- Another example is the quota on U.S. imports of sugar, which is still in place despite calls for its removal.

Import Quotas in a Small Country

Effect of Import Quota for a Small Economy:

FIGURE 8-9 Quota for a Small Country



Under free trade, the Foreign export supply curve is horizontal at the world price P^W, and the free-trade equilibrium is at point B with imports of M_1. Applying an import quota of $M_2 < M_1$ leads to the vertical export supply curve, with the equilibrium at point C.

The quota increases the import price from P^W to P_2. There would be the same impact on price and quantities if instead of the quota, a tariff of $t = P_2 - P^W$ had been used.

For every level of the import quota, there is an **equivalent import tariff** that would lead to the same Home price and quantity of imports.

Import Quotas in a Small Country

Effect of Import Quota for a Small Economy:

- Whoever is actually importing the good will be able to earn the difference between the world price P^W and the higher Home price P_2 by selling the imports in the Home market.
- We call the difference between these two prices the rent associated with the quota, and hence the area c represents the total **quota rents**.
- Next we examine the four possible ways that these quota rents can be allocated.

Import Quotas, cont'

1. *Giving the Quota to Home Firms*: **Quota licenses** (i.e., permits to import the quantity allowed under the quota system) can be given to Home firms. With home firms earning the rents c, the net effect of the quota on Home welfare is:

   - Fall in consumer surplus: - (a + b + c + d)
   - Rise in producer surplus: + a
   - Quota rents earned at Home: + c
   - Net effect on Home welfare: - (b + d)

Import Quotas, cont'

2 If licenses for the imported chemicals are allocated in proportion to each firm's production of batteries in the previous years, then the Home firms will likely produce more batteries than they can sell (and at lower quality) just to obtain the import licenses for the following year.
Alternatively, firms might engage in bribery or other lobbying activities to obtain the licenses. These kinds of inefficient activities done to obtain quota licenses are called **rent seeking**.

3 **Auctioning the Quota** A third possibility for allocating the rents that come from the quota is for the government of the importing country to auction off the quota licenses.
In a well-organized, competitive auction, the revenue collected should exactly equal the value of the rents, so that area c would be earned by the Home government.

Import Quotas, cont'

4 **"Voluntary" Export Restraint** The final possibility for allocating quota rents is for the government of the importing country to give authority for implementing the quota to the government of the *exporting country*.
Because the exporting country allocates the quota among its own producers, this is sometimes called a **"voluntary" export restraint** (**VER**), or a **"voluntary" restraint agreement** (**VRA**). In the 1980s the United States used this type of arrangement to restrict Japanese automobile imports. In this case, the quota rents are earned by foreign producers.

Summary

Key Points:

1. The government of a country can use laws and regulations, called "trade policies", to affect international trade flows. An import tariff, which is a tax at the border, is the most commonly used trade policy.

2. The rules governing trade policies in most countries are outlined by the General Agreement on Tariffs and Trade (GATT), an international legal convention adopted after World War II to promote increased international trade. Since 1995 the new name for the GATT is the World Trade Organization (WTO).

3. In a small country, the quantity of imports demanded is assumed to be very small compared with the total world market. For this reason, the importer faces a fixed world price. In that case, the price faced by consumers and producers in the importing country will rise by the full amount of the tariff.

Summary

Key Points:

4. The use of a tariff by a small importing country always leads to a net loss in welfare. We call that loss the "deadweight loss".

5. In a large country, the decrease in imports demanded due to the tariff causes foreign exporters to lower their prices. Consumer and producer prices in the importing country still go up, since these prices include the tariff, but they rise by less than the full amount of the tariff (since the exporter price falls).

6. The use of a tariff for a large country can lead to a net gain in welfare because the price charged by the exporter has fallen; this is a terms-of-trade gain for the importer.

Summary

Key Points:

7. The "optimal tariff" is the tariff amount that maximizes welfare for the importer. For a small country, the optimal tariff is zero since any tariff leads to a net loss. For a large country, however, the optimal tariff is positive.

8. The formula for the optimal tariff states that it depends inversely on the foreign export supply elasticity. If the foreign export supply elasticity is high, then the optimal tariff is low, but if the foreign export supply elasticity is low, then the optimal tariff is high.

9. "Import quotas" restrict the quantity of a particular import, thereby increasing the domestic price, increasing domestic production, and creating a benefit for those who are allowed to import the quantity allotted. These benefits are called "quota rents".

Summary

Key Points:

10. Assuming perfectly competitive markets for goods, quotas are similar to tariffs since the restriction in the amount imported leads to a higher domestic price. However, the welfare implications of quotas are different from those of tariffs depending on who earns the quota rents. These rents might be earned by firms in the importing country (if they have the licenses to import the good), or by firms in the exporting country (if the foreign government administers the quota), or by the government in the importing country (if it auctions off the quota licenses). The last case is most similar to a tariff, since the importing government earns the revenue.